A MINIMAL POINT OF THE SUPPORT OF A QUASI-COHERENT MODULE IS ASSOCIATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- A nonzero module over a Noetherian local ring has an associated prime. Algebra, Lemma algebra-lemma-ass-zero, used at proof lines 232-235. Makes Ass(F_x) nonempty.
- Associated primes lie in the support. Algebra, Lemma algebra-lemma-ass-support, used at proof lines 238-240. Places the associated prime inside the support of F_x.

Source reference. Original lemma 0586: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6769–6769. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0587: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6770–6770. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a locally Noetherian scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $x \in Supp(\mathcal{F})$ be a point in the support of $\mathcal{F}$ which is not a specialization of another point of $Supp(\mathcal{F})$. Then $x \in Ass(\mathcal{F})$. In particular, any generic point of an irreducible component of X is an associated point of X.*

Proof. Since $x \in \mathrm{Supp}(\mathcal{F})$ the module $\mathcal{F}_x$ is not zero. Hence $\mathrm{Ass}(\mathcal{F}_x) \subset \mathrm{Spec}(\mathcal{O}_{X,x})$ is nonempty by Algebra, Lemma 0587. On the other hand, by assumption $\mathrm{Supp}(\mathcal{F}_x) = \{\mathfrak{m}_x\}$. Since $\mathrm{Ass}(\mathcal{F}_x) \subset \mathrm{Supp}(\mathcal{F}_x)$ (Algebra, Lemma 0586) we see that $\mathfrak{m}_x$ is associated to $\mathcal{F}_x$ and we win. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.